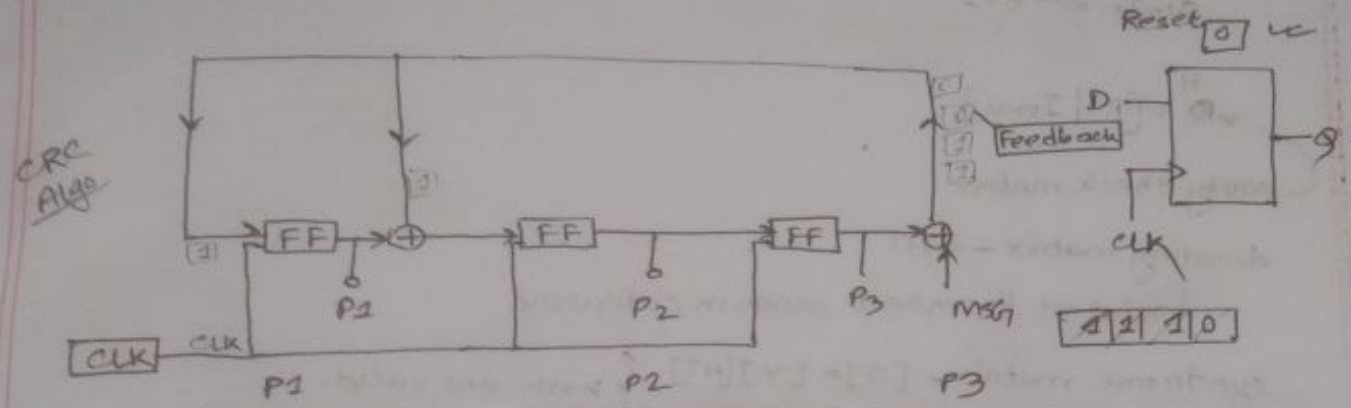
CSE-311

03/03/2026

Designing of Cyclic Encoder

(7,4) ; $G(x) = x^3 + x + 1$



P1

prev	curr
	0
0	1
1	1
1	0
0	0

P2

prev	curr
	0
0	1
1	0
0	1
1	0

P3

prev	curr
	0
0	0
0	1
1	0
0	1

CLK	D	P1	P2	P3
↑		0	0	0
↑	1	1	1	0
↑	1	1	0	1
↑	1	0	1	0
↑	0	0	0	1

MSB

P3 P2 P1

1	0	0

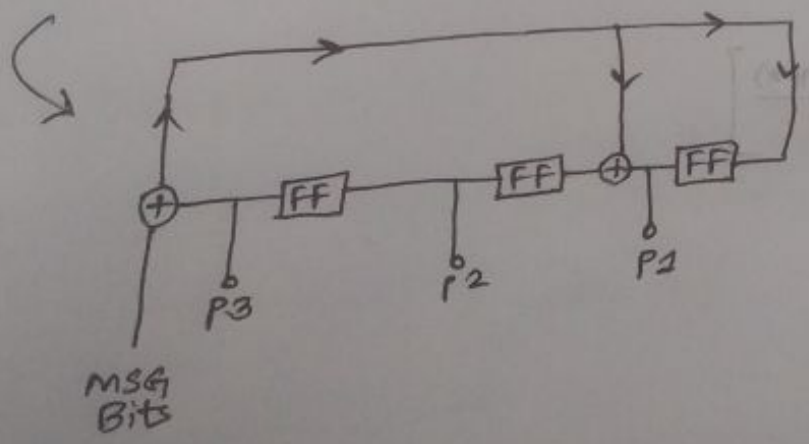
Feedback

P3 ⊕ MSG

H/W implementation

$$P(x) = \text{Rem}\left[\frac{x^{n-k} m(x)}{g(x)}\right]$$

CRC (Linear Feedback Shift Register) Algorithm



⇒ mirroned

↑ ⇒ left shift

P3 P2 P1 ⇒ remainder

⊛ Systematic (7,4)

$G(x) = x^3 + x + 1$

$\overset{H}{\cancel{R}} = [P^T \mid I_{n-k}]$

parity check matrix

decoding matrix $\rightarrow H^T$

↳ detect & correct error in codeword

syndrome matrix, $[S] = [Y][H^T]$

$S = Rem\left[\frac{y(x)}{g(x)}\right]$ } both are valid.

$c(x) = g(x) \cdot m(x) \Rightarrow c(x)$ should be divisible by $g(x)$

↳ remainder error

⇒ match syndrome with rows of decoding matrix to find erroneous position

<u>CRC</u> ⇒ Cyclic Redundancy Check

divisor ⇒ generator in binary

⊛ message ⇒ 11001001 received ⇒ 11011001011

CRC polynomial ⇒ x^3+1 → $\boxed{\text{Divisor} = 1001}$

or

generator polynomial

$n-k=3$ $Rem = \left[\frac{x^{n-k} m(x)}{g(x)}\right]$

shift left 3 times

```
        10110
1001 | 11001001 | 000
       1001
       001110
       001001
         01010
         01001
          00011
```

$\boxed{011}$

$Tx = \underbrace{11001001}_{data}\ \underbrace{011}_{CRC}$

Decoding

```
1001 | 11011 001011 |
       1001
        1001
        1001
           001011
              1001
                10
```

→ error

$\boxed{\text{no connection}}$

CRC ⇒ can only detect the error but not the position

use syndrome.. for position.

Convolutional Code

→ time dependancy → clk..

⇒ process data continuously, not in fixed-sized blocks

output sequence is the convolution of the input sequence with generator

constraint length (K) = # memory elements + input size (→ at a time → 1)

= m+1

→ # bits involved in producing the o/p

symbols

dimension (n,k) ⇒ (2,1)

→ only represents the o/p bits

→ # i/p at a time

→ # o/p at a time

representation ⇒ (n,k,K)

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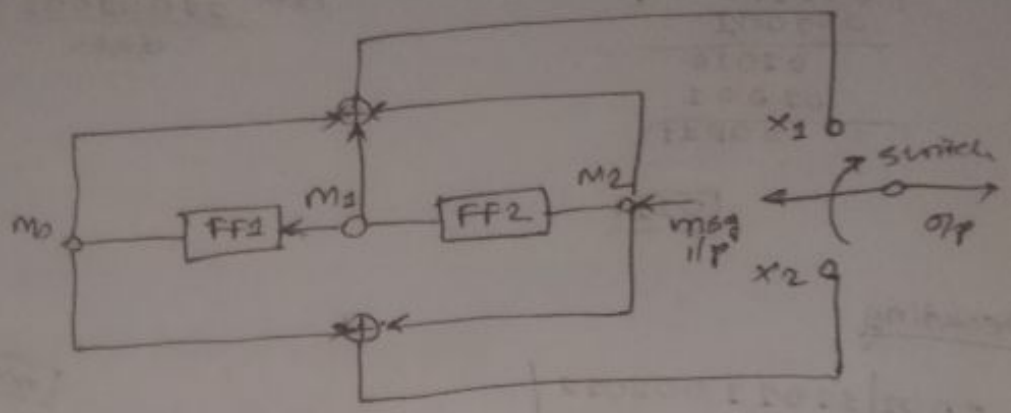
code rate, $r = \frac{k}{n}$ → at a time i/p
→ at a time o/p

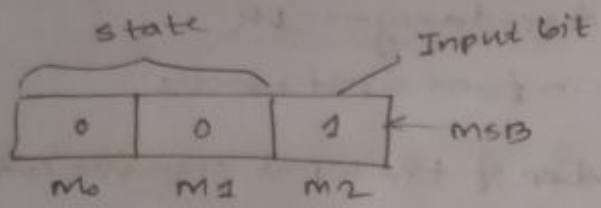


states

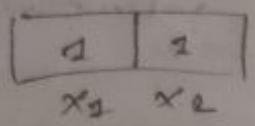
M_0	M_1	
0	0	→ a
0	1	→ b
1	0	→ c
1	1	→ d

$2^2 = 4$

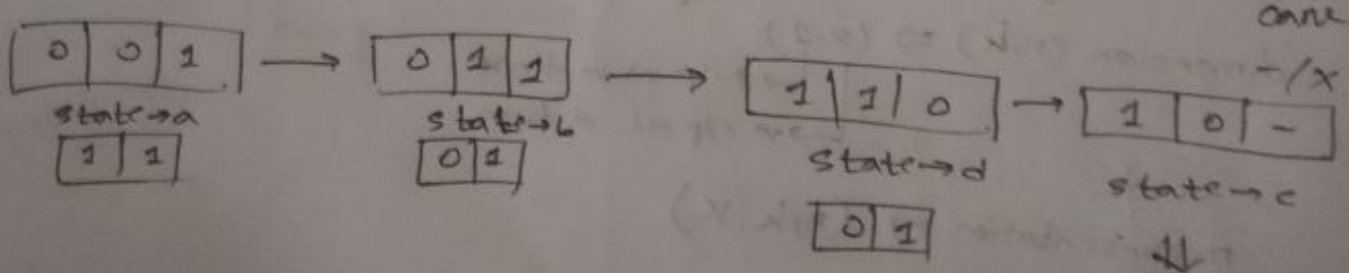
convolutional code
↳ o/p sequence on o/p
$\boxed{\text{not codeword}}$



Message: [1 1 0]



CLK pulse ⇒ $M_0 \leftarrow M_1$
$M_1 \leftarrow M_2$



⇓
o/p not needed

$\boxed{110101}$ → output seq

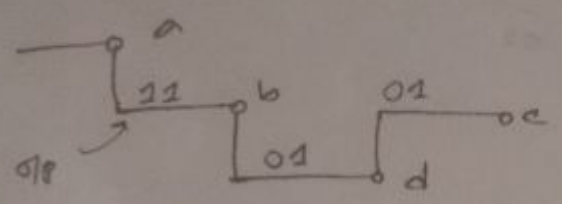
Code Tree

i/p = 1 → step down
i/p = 0 → step up

step → horizontal

initially always state = a

reset ⇒ 00



Code Trellis and State Diagram

truth table ⇒ i/p: M_0, M_1, M_2

→ state: M_0M_1

CLK

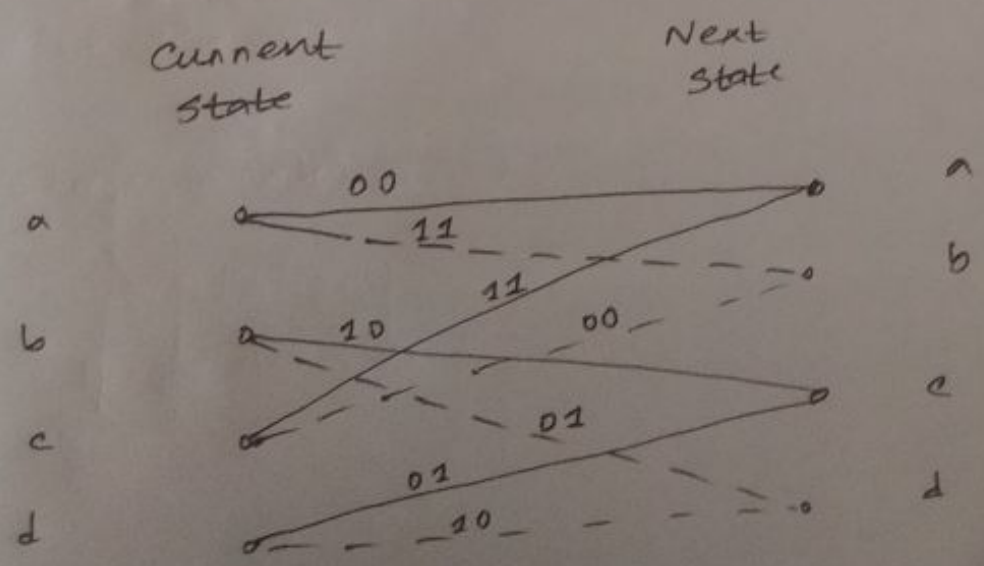
→ state: M_1M_2

M_0	M_1	M_2	x_1	x_2	current state	next state
0	0	0	0	0	a	a
0	0	1	1	1	a	b
0	1	0	1	0	b	c
0	1	1	0	1	b	d
1	0	0	1	1	c	a
1	0	1	0	0	c	b
1	1	0	0	1	d	c
1	1	1	1	0	d	d

code trellis → diagram of current & next states

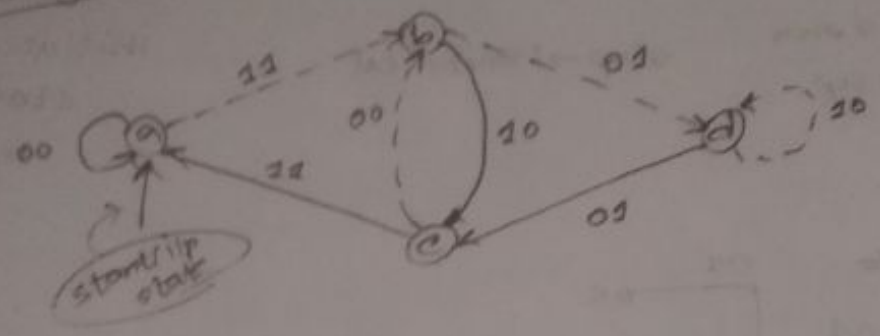
i/p = 0 → solid line
i/p = 1 → dotted line

may use the inverse



State diagram

i/p ⇒ 1 bit
⇒ 2 transitions from all states



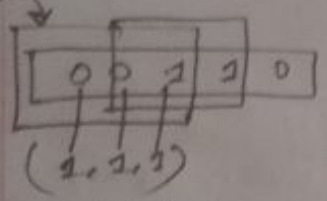
why convolutional?
sliding window from right to left

alternative ckt repn:

generator seq $g_1(x) = (1,1,1) \rightarrow x_1 =$

$g_2(x) = (1,0,1) \rightarrow x_2 =$

sliding window



weighted sum / XOR

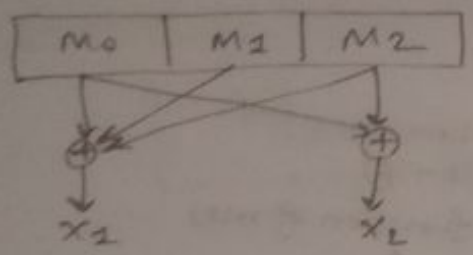
$x_1 = 1$

$x_2 = 1$

sliding operation ⇒ convolution

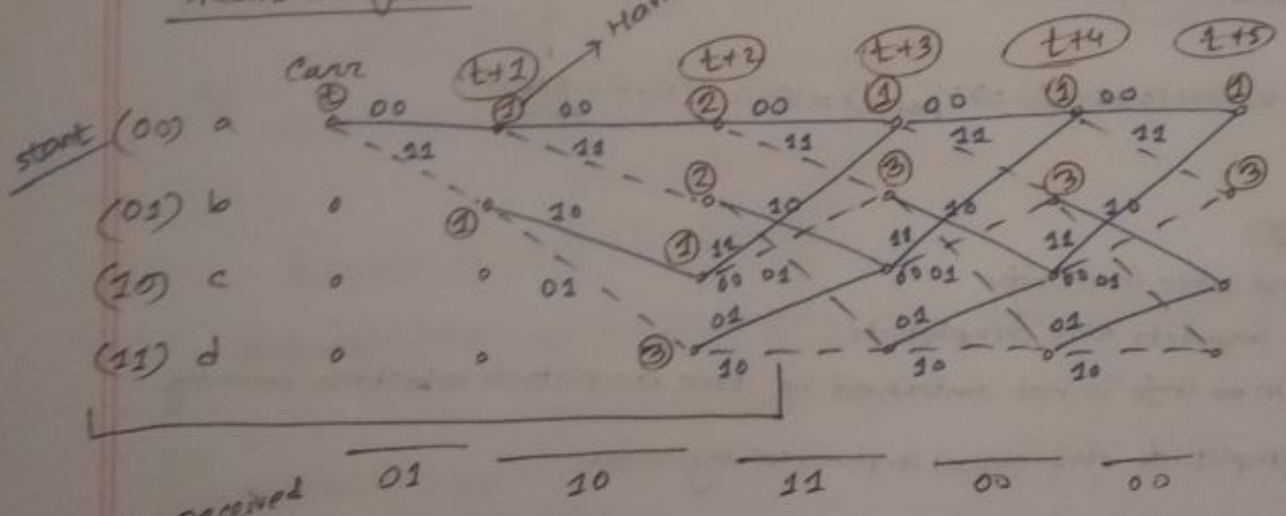
Viterbi Algorithm → indirect
does not detect & correct directly.
→ find the most likely transmitted bit sequence.

closest sequence



received seq
01 10 11 00 00
↳ 10bit, at a time 1 i/p, 2 o/p
5 CLK pulses needed
5 next states

Trellis diagram



a → b → c ⇒ closest seq

wt: 0 → best sequence

11 10 → emanneous bit: 1

1
1 1 0

CSE-311

07/02/2026

(AM) Amplitude Modulation
→ modifies the amplitude

Angle Modulation

→ frequency or phase of carrier signal changes.

(2) types
→ frequency modulation
→ phase modulation

FM

$e(t) = V_c \cos(\omega_c t + \phi) \Rightarrow Y_{FM}(t) = V_c \cos(\omega(t) t + \phi)$

→ time dependent

→ function of $m(t)$

carrier signal

$Y_c(t) = A_c \cos(\omega_c t)$

A_c → fixed

$\omega(t) \Rightarrow$ phase

PM

$e(t) = V_c \cos(\omega_c t + \phi) \Rightarrow Y_{PM}(t) = V_c \cos(\omega_c t + \phi(t))$

Advantages

(*) improved noise immunity

noise impacts the amplitude

FM, PM ⇒ info is not contained in the amplitude ⇒ better immunity

↳ if amplitude changes ⇒ impacted by noise

(*) constant amplitude reduces power loss due to non-linearities in amplifiers.

↳ efficient → power consumption ↓

linear amplifier ⇒ consumes a lot of power

(AM) → non-linear ⇒ distortion occurs as amp contains info.

(*) better signal quality

[capture effect] → 2 FM signals can be differentiated based on power

AM → mixes both

- higher efficiency in BW ⇒ narrowband FM
- less susceptible to signal fading & amp variations ⇒ error prone

AM → amp fluctuates

Instantaneous Angular Frequency, $\omega_i(t)$

$\omega = 2\pi f$

↳ frequency

→ angular frequency

$\omega_i(t) = \frac{d}{dt}\theta(t)$ → instantaneous phase

Carrier Signal

$y_c(t) = A_c \cos(\omega_c t)$ → $\theta(t)$

$k_p \Rightarrow$ rad/Volt

PM

$\theta(t) = \omega_c t + \underbrace{k_p m(t)}_{\text{linear rel}^n \text{ with } m(t)}$ → instantaneous phase

k_p → phase sensitivity constant

$y_{pm}(t) = A_c \cos[\omega_c t + k_p m(t)]$

instantaneous angular frequency, $\omega_i(t) = \omega_c + k_p \frac{d}{dt} m(t)$

FM

(Hz) $\omega_i(t) = \omega_c + k_f m(t)$ — Volt

k_f → frequency sensitivity

k_f → Hz/Volt

$$\begin{cases} FM = PM \text{ of } \int m(t)\,dt \longrightarrow y_{pm}(t) = A_c \cos\left[\omega_c t + k_p \int m(t)\,dt\right] \\ PM = FM \text{ of } \frac{dm(t)}{dt} \end{cases}$$

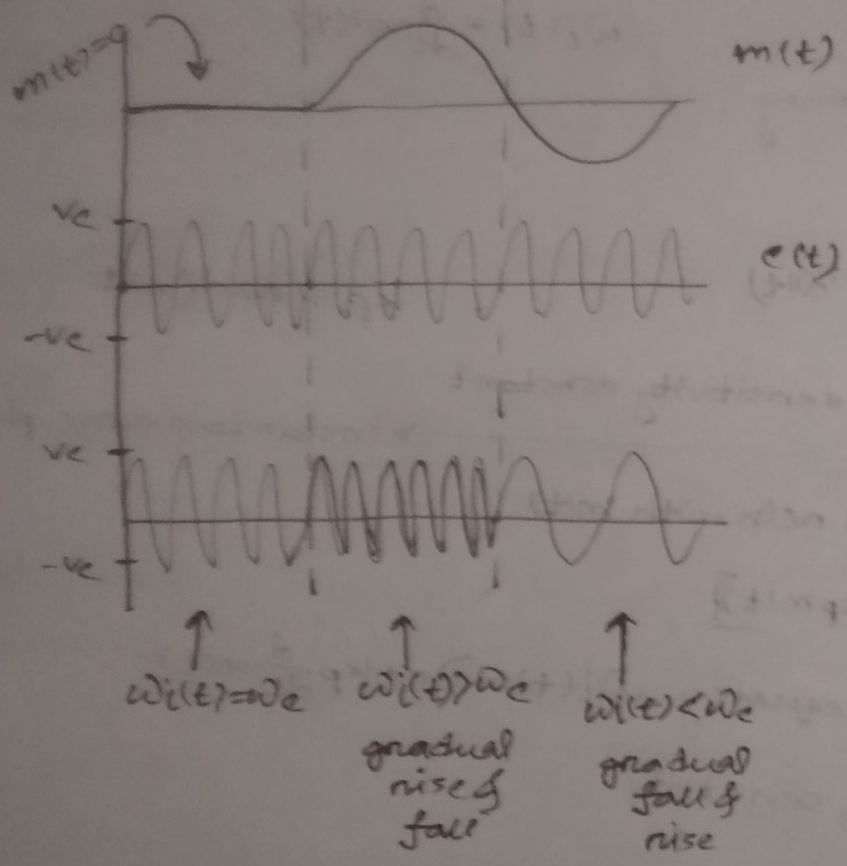
Frequency Modulation

$$\theta(t) = \omega_c t + 2\pi k_F \int_0^t m(t)\,dt$$

carrier → High freq

$m(t) \longrightarrow$ Integrator $\longrightarrow$ Phase Modulator $\longrightarrow$ FM wave

Waveforms



HW
waveforms of PM

square wave,
↳ increase, decrease → ×